

Wold-type decomposition for symplectic maps: wandering space and the J -shift

Abstract

Let H be a real Hilbert space with inner product g and a compatible complex structure J ($J^2 = -I$, $J^* = -J$), giving the strong symplectic form $\Omega(x, y) := g(Jx, y)$. Let T be a bounded linear map on H preserving Ω . We give a full Wold-type decomposition of H relative to T in two regimes. When T is additionally a metric isometry, the classical Wold–von Neumann decomposition applies verbatim, and we show that Ω , restricted to the shift part, is a block-Toeplitz form whose symbol $\Phi_k = P_N J T^k|_N$ (the *J-shift symbol*) encodes exactly how J interacts with the shift; Φ_0 is the compression of J to the wandering space. When T is only Ω -preserving (not an isometry), we show that the symplectic wandering filtration, its pairwise Ω -orthogonality, and the residual (“unitary”) part all survive unconditionally, while the passage to a convergent Hilbert-space direct sum genuinely requires an extra hypothesis; we make this precise via the metric wandering space and the class $\mathcal{D}$ of left-invertible operators of Shimorin and Manolescu, and verify it on an explicit non-isometric example.

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1 Setup and notation

- (H, g) is a real Hilbert space, inner product g , norm $\|\cdot\|$.
- $J : H \rightarrow H$ is bounded linear with $J^2 = -I$ and $J^* = -J$. Equivalently ($J^*J = (-J)J = -J^2 = I$) J is g -orthogonal: $\|Jx\| = \|x\|$ and $g(Jx, Jy) = g(x, y)$ for all x, y .
- $\Omega(x, y) := g(Jx, y)$ is the *symplectic form*. It is bounded ($|\Omega(x, y)| \leq \|x\|\|y\|$), antisymmetric, and *strong*: the map $\flat : x \mapsto \Omega(x, \cdot)$ is a Banach-space isomorphism $H \rightarrow H^*$ (composition of the Riesz isomorphism with the isomorphism J).

- $T : H \rightarrow H$ is bounded linear and Ω -preserving: $\Omega(Tx, Ty) = \Omega(x, y)$ for all x, y , equivalently $T^*JT = J$.

Lemma 1.1 (Basic identities). $\|Jx\| = \|x\|$ for all x ; and $\Omega(Tx, Ty) = \Omega(x, y)$ for all x, y if and only if $T^*JT = J$.

Proof. $J^*J = (-J)J = -J^2 = I$, so $\|Jx\|^2 = g(Jx, Jx) = g(x, J^*Jx) = g(x, x) = \|x\|^2$. For the second claim, $\Omega(Tx, Ty) = g(JTx, Ty) = g(T^*JT x, y)$ and $\Omega(x, y) = g(Jx, y)$; the two agree for all x, y iff $T^*JT = J$. $\square$

Lemma 1.2 (Metric/symplectic complement duality). For every closed subspace $M \subseteq H$,

$$M^{\perp\Omega} := \{v \in H : \Omega(v, M) = 0\} = J(M^{\perp g}), \quad M^{\perp g} := \{v \in H : g(v, M) = 0\}.$$

Proof. $v \in M^{\perp\Omega} \iff g(Jv, m) = 0 \ \forall m \in M \iff Jv \in M^{\perp g} \iff v \in J^{-1}(M^{\perp g})$. Since $J^2 = -I$, $J^{-1} = -J$, and for a linear subspace S we have $-S = S$, so $-J(S) = J(-S) = J(S)$. Hence $J^{-1}(M^{\perp g}) = J(M^{\perp g})$, giving the claim. $\square$

This holds for *any* closed M and uses nothing about T : it is the bridge between the metric (g) and symplectic (Ω) notions of complement.

The prototype to keep in mind throughout: $H = \bigoplus_{n \geq 1} \mathbb{R}^{2d}$, $J = \bigoplus_n J_0$ (J_0 the standard symplectic unit on $\mathbb{R}^{2d}$), and $T = S$ the right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$.

2 The case of a symplectic isometry

Throughout this section T is a **symplectic isometry**:

$$\|Tx\| = \|x\| \quad \text{and} \quad \Omega(Tx, Ty) = \Omega(x, y) \quad \text{for all } x, y \in H,$$

equivalently $T^*T = I$ and $T^*JT = J$.

2.1 The metric Wold decomposition of T

This is the classical Wold–von Neumann theorem (Halmos’ proof), stated so that Section 2.2 can refer to each step.

Range closedness. $R_1 := TH$ is closed: if $Tx_n \rightarrow z$ then $\|x_n - x_m\| = \|Tx_n - Tx_m\| \rightarrow 0$ (isometry), so $x_n \rightarrow x$ for some $x \in H$, and $Tx = z$ by continuity. The same argument applied to the isometry T^n shows $R_n := T^n H$ is closed for every $n \geq 0$, and $R_{n+1} = T(R_n) \subseteq R_n$ (since $TH \subseteq H$ implies $T^n(TH) \subseteq T^n(H)$): a decreasing chain of closed subspaces.

Wandering space. Define

$$N := H \ominus TH = (TH)^{\perp g} = R_1^{\perp g}, \quad N_n := T^n N \ (n \geq 0), \quad N_0 = N.$$

Proposition 2.1 (W1–W8). (W1) $T^n : N \rightarrow N_n$ is a linear isometry onto N_n .

(W2) $N_n \perp_g R_{n+1}$.

(W3) $R_n = R_{n+1} \oplus_g N_n$.

(W4) $N_k \perp_g N_l$ for $k \neq l$.

(W5) With $H_s := \overline{\text{span}}(\bigcup_{n \geq 0} N_n)$ and $H_\infty := \bigcap_{n \geq 0} R_n$, the map $\Psi : \ell^2(N) \rightarrow H_s$, $\Psi((a_n)_{n \geq 0}) = \sum_{n \geq 0} T^n a_n$, is a Hilbert-space isomorphism.

(W6) $H = H_s \oplus_g H_\infty$.

(W7) $T|_{H_\infty}$ is a surjective isometry (unitary) of H_∞ .

(W8) $T|_{H_s}$ is the unilateral shift of multiplicity N : $T \circ \Psi = \Psi \circ \Sigma$ where $\Sigma(a_0, a_1, \dots) = (0, a_0, a_1, \dots)$.

Proof. (W1) T^n is an isometry (a composition of isometries), hence injective, and it maps the closed set N onto the closed set $N_n := T^n N$ (closed by the same closed-range argument as above); by definition $\|T^n a\| = \|a\|$ for $a \in N$.

(W2) Let $a \in N$, $r \in R_{n+1} = T^n(TH)$, say $r = T^n(Ts)$. Then $g(T^n a, r) = g(T^n a, T^n(Ts)) = g(a, Ts) = 0$ (isometry preserves g ; $a \in N = (TH)^{\perp_g}$, $Ts \in TH$).

(W3) Given $h \in H$, write $h = w + a$ with $w \in TH$, $a \in N$ (projection theorem, TH closed). Apply T^n : $T^n h = T^n w + T^n a$, with $T^n w \in T^n(TH) = R_{n+1}$ and $T^n a \in N_n$. So $R_n \subseteq R_{n+1} + N_n$; combined with (W2) this is the orthogonal decomposition $R_n = R_{n+1} \oplus_g N_n$.

(W4) If $l > k$ then $N_l \subseteq R_l \subseteq R_{k+1}$ (decreasing chain), and $N_k \perp_g R_{k+1}$ by (W2); hence $N_k \perp_g N_l$.

(W5) Ψ is linear and onto by definition of H_s , and it is an isometry by the Pythagorean theorem: $\|\Psi(a)\|^2 = \sum_n \|T^n a_n\|^2 = \sum_n \|a_n\|^2$ (using (W1), (W4)).

(W6) *Orthogonality*: for $a \in N_n$ and $h \in H_\infty \subseteq R_{n+1}$, (W2) gives $a \perp h$; this holds for every n , so $H_s \perp_g H_\infty$. *Spanning*: fix $h \in H$. Iterating (W3), for every n there are unique $a_0, \dots, a_{n-1} \in N$ and $h_n \in R_n$ with $h = h_n + \sum_{k < n} a_k$, and uniqueness of the orthogonal decomposition (W3) shows the a_k do not change as n grows. Since the a_k are pairwise orthogonal, $\sum_{k < n} \|a_k\|^2 = \|h\|^2 - \|h_n\|^2 \leq \|h\|^2$, so $\sum_k \|a_k\|^2 < \infty$; put $a := \sum_k T^k a_k \in H_s$. Then $h_n = h - \sum_{k < n} a_k \rightarrow h - a =: h_\infty$. For fixed m and all $n \geq m$, $h_n \in R_n \subseteq R_m$ (closed), so $h_\infty \in R_m$; as m was arbitrary, $h_\infty \in \bigcap_m R_m = H_\infty$. Thus $h = a + h_\infty \in H_s + H_\infty$.

(W7) *Invariance*: $T(H_\infty) = T(\bigcap_n R_n) \subseteq \bigcap_n T(R_n) = \bigcap_n R_{n+1} = H_\infty$. *Onto*: let $h \in H_\infty$. Since $h \in R_1 = TH$, there is (by global injectivity of T) a unique $x \in H$ with $Tx = h$. For each n , $h \in R_{n+1} = T(R_n)$, so $h = Ty$ for some $y \in R_n$; by uniqueness of the preimage, $y = x$, hence $x \in R_n$. As n was arbitrary, $x \in \bigcap_n R_n = H_\infty$ and $Tx = h$. So $T(H_\infty) = H_\infty$, and $T|_{H_\infty}$, being an isometry onto itself, is unitary.

(W8) $\Psi(0, a_0, a_1, \dots) = \sum_{n \geq 1} T^n a_{n-1} = T(\sum_{n \geq 0} T^n a_n) = T\Psi(a_0, a_1, \dots)$, i.e. $T \circ \Psi = \Psi \circ \Sigma$. $\square$

Theorem 2.2 (Wold decomposition). $H = H_s \oplus_g H_\infty$, both summands reduce T , $T|_{H_\infty}$ is unitary, and $T|_{H_s} \cong \Sigma \otimes I_N$ is a unilateral shift with wandering space $N = H \ominus TH$.

Nothing here has used Ω or J yet. They enter now.

2.2 Where J enters: the symplectic defect and the J -shift symbol

By Lemma 1.2 with $M = R_1 = TH$,

$$R_1^{\perp \Omega} = J(R_1^{\perp g}) = J(N).$$

More generally, since (W3)/(W4) give $R_n^{\perp g} = \bigoplus_{k < n} N_k$ (finite g -orthogonal sum, complement of R_n in H), Lemma 1.2 gives, for every n ,

$$R_n^{\perp \Omega} = J\left(\bigoplus_{k < n} N_k\right) = \bigoplus_{k < n} J(N_k),$$

the last equality because J preserves g , hence preserves orthogonality. So the symplectic wandering space $R_1^{\perp \Omega}$ of T is $J(N)$, not N itself: the metric wandering space and the symplectic defect space of TH coincide up to the fixed rotation J .

For $a, b \in N$ and $k \geq 0$ define the bounded bilinear form

$$\varphi_k(a, b) := \Omega(T^k a, b) = g(JT^k a, b),$$

equivalently the bounded operator

$$\Phi_k := P_N J T^k|_N : N \rightarrow N, \quad g(\Phi_k a, b) = \varphi_k(a, b),$$

where P_N is the g -orthogonal projection of H onto N . Since J and T^k are both g -isometries, $\|\Phi_k\| \leq 1$ for every $k \geq 0$; in particular $\Phi_0 = P_N J|_N$ is the *compression of J to the wandering space*.

Lemma 2.3 (Reduction). *For $a, b \in N$ and $n \geq m \geq 0$, $\Omega(T^n a, T^m b) = \varphi_{n-m}(a, b)$.*

Proof. Preservation of Ω by T iterated m times gives $\Omega(T^m x, T^m y) = \Omega(x, y)$ for all $x, y \in H$. Take $x = T^{n-m} a$, $y = b$: $\Omega(T^n a, T^m b) = \Omega(T^m(T^{n-m} a), T^m b) = \Omega(T^{n-m} a, b) = \varphi_{n-m}(a, b)$. $\square$

By antisymmetry of Ω , for $n < m$: $\Omega(T^n a, T^m b) = -\Omega(T^m b, T^n a) = -\varphi_{m-n}(b, a)$.

Theorem 2.4 (The J -shift form of Ω). *Under the isomorphism $\Psi : \ell^2(N) \rightarrow H_s$ of (W5), for $a = (a_n)_{n \geq 0}$, $b = (b_n)_{n \geq 0} \in \ell^2(N)$,*

$$\Omega(\Psi a, \Psi b) = \sum_{k \geq 0} \sum_{m \geq 0} \varphi_k(a_{m+k}, b_m) - \sum_{k \geq 1} \sum_{n \geq 0} \varphi_k(b_{n+k}, a_n),$$

where both double series are the (well-defined) values of $\lim_{N, M \rightarrow \infty} \sum_{n=0}^N \sum_{m=0}^M \Omega(T^n a_n, T^m b_m)$, which exists and equals $\Omega(\Psi a, \Psi b)$ because Ω is a bounded (hence jointly continuous) bilinear form and $\Psi a = \lim_N \sum_{n \leq N} T^n a_n$, $\Psi b = \lim_M \sum_{m \leq M} T^m b_m$ in norm.

Proof. Split the finite double sum $\sum_{n \leq N, m \leq M} \Omega(T^n a_n, T^m b_m)$ into $n \geq m$ and $n < m$ and apply Lemma 2.3 to each term; let $N, M \rightarrow \infty$ and substitute $k = n - m$ (resp. $k = m - n$). $\square$

We call the operator sequence $(\Phi_k)_{k \geq 0}$, $\Phi_k = P_N J T^k|_N$, the **J -shift symbol of T** : it is exactly the extra data (beyond the shift Σ itself) needed to reconstruct Ω on the pure-shift part H_s . Theorem 2.4 says $\Omega|_{H_s}$ is a *block-Toeplitz form* with that symbol.

Corollary 2.5. $\varphi_0(a, b) = \Omega(a, b)$, i.e. $\Omega|_N$ is recovered from the symbol at $k = 0$.

Example 2.6 (The prototype has trivial symbol). For $H = \bigoplus_{n \geq 1} \mathbb{R}^{2d}$, $J = \bigoplus_n J_0$, $T = S$ the right shift: N = the first block, N_n = the $(n+1)$ -st block, $H_\infty = \bigcap_n S^n H = \{0\}$ (S is a *pure* shift). For $a \in N$, $S^k a$ lives entirely in block $k+1$; since J is block-diagonal, $J S^k a$ also lives entirely in block $k+1$, so $P_N(J S^k a) = 0$ for $k \geq 1$ and $P_N(J a) = J_0 a$ for $k = 0$. Hence

$$\Phi_0 = J_0, \quad \Phi_k = 0 \quad (k \geq 1) :$$

the symbol is trivial, and Theorem 2.4 collapses to $\Omega(\Psi a, \Psi b) = \sum_{n \geq 0} g(J_0 a_n, b_n)$ — the blocks are Ω -orthogonal to each other, not just g -orthogonal. This is special to J being *adapted* to the shift's block structure. In general $\Phi_k \neq 0$ for $k \geq 1$ measures the failure of J to be adapted to the filtration $R_0 \supseteq R_1 \supseteq \dots$ generated by T .

Remark 2.7. Only §2 needs $T^* T = I$: it gives the Pythagorean identity that makes H_s an orthogonal (not merely topological) direct sum and makes Φ_k uniformly bounded. If T commutes with J (complex-linear), Ω -preservation is *automatic* from isometry alone: $\Omega(Tx, Ty) = g(JTx, Ty) = g(TJx, Ty) = g(Jx, y) = \Omega(x, y)$. The interesting case, where the symbol $(\Phi_k)_{k \geq 1}$ can be nonzero, is precisely when T is a *real* (non-complex-linear) symplectic isometry: from $T^* J T = J$ and $T T^* = P_{TH}$ one gets $TJ = P_{TH}(JT)$, i.e. $JT - TJ$ takes values in $N = (TH)^\perp_g$ — a defect operator $\delta(x) := JT x - TJ x \in N$ measuring the noncommutativity that feeds the nonzero symbol terms. Finally, H_∞ is again a symplectic Hilbert space and $T|_{H_\infty}$ is a **symplectic unitary** (metrically unitary and Ω -preserving).

3 The general case: T need not be an isometry

We now drop the isometry hypothesis entirely: T is bounded and Ω -preserving only.

Example 3.1 (A genuinely non-isometric symplectic map). Take $d = 1$, $J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ on $\mathbb{R}^2$ (so $\Omega_0(u, v) = u_1 v_2 - u_2 v_1$), fix $\lambda > 1$, and let $T_0 := \text{diag}(\lambda, \lambda^{-1})$. Since $\det T_0 = 1$, T_0 is symplectic ($\Omega_0(T_0 u, T_0 v) = \Omega_0(u, v)$) but not orthogonal ($\|T_0 e_1\| = \lambda \neq 1$). Put $H = \bigoplus_{n \geq 1} \mathbb{R}^2$, $J = \bigoplus_n J_0$, and

$$T(x_1, x_2, x_3, \dots) := (0, T_0 x_1, T_0 x_2, \dots)$$

(a “matrix-weighted shift”). Then $\Omega(Tx, Ty) = \sum_n g(J_0 T_0 x_n, T_0 y_n) = \sum_n \Omega_0(T_0 x_n, T_0 y_n) = \sum_n \Omega_0(x_n, y_n) = \Omega(x, y)$: T preserves Ω . But $\|Tx\|^2 = \sum_n \|T_0 x_n\|^2 \neq \sum_n \|x_n\|^2 = \|x\|^2$ in general: T is not an isometry. We compute everything on this example as we go.

3.1 Three facts that use no isometry

The following are proved by a direct dilation/complement argument (Step 1–3 of the one-step extension construction; see Remark 3.5 for provenance) using only boundedness and Ω -preservation.

Fact 3.2 (Bounded below). $\|Tx\| \geq c\|x\|$ for all x , with $c := 1/\|T\|$. Hence T is injective and $R_1 := TH$ is closed.

Proof. By Lemma 1.1, $\|x\| = \|Jx\| = \|T^* JTx\| \leq \|T\| \|JTx\| = \|T\| \|Tx\|$, so $\|Tx\| \geq \|x\|/\|T\|$. If $Tx_n \rightarrow z$ then (x_n) is Cauchy (by the bound applied to $x_n - x_m$), so $x_n \rightarrow x$ and $Tx = z$ by continuity; hence R_1 is closed. $\square$

Fact 3.3 (Strength is inherited). $\Omega|_{R_1}$ is again a strong symplectic form.

Fact 3.4 (One-step complement). $H = R_1 \oplus N_0$ (a topological, bounded-idempotent direct sum — not g -orthogonal), where $N_0 := R_1^{\perp \Omega} = \{v : \Omega(v, R_1) = 0\}$, $\Omega(R_1, N_0) = 0$, $(N_0, \Omega|_{N_0})$ is strong symplectic, and $T : H \rightarrow R_1$ is a bijective, bounded-both-ways symplectomorphism.

Remark 3.5. Facts 3.2–3.4 are the symplectic analogue of dilating a Hilbert-space isometry to a unitary; N_0 plays the role of the defect space, and $E := B^{-1}P_{R_1}J$ (with $B := P_{R_1}J|_{R_1}$) is the bounded idempotent onto R_1 with kernel N_0 .

In Example 3.1: $R_1 = \{(0, y_1, y_2, \dots)\}$ (since T_0 is invertible), and $N_0 = R_1^{\perp \Omega} = J(R_1^{\perp g})$ (Lemma 1.2, no isometry needed) equals $J_0(\text{first block}) = \text{first block}$, since J_0 maps the first $\mathbb{R}^2$ block to itself.

3.2 The Ω -wandering filtration

Define $R_n := T^n H$ ($R_0 = H$).

Proposition 3.6. *For every $n \geq 0$: R_n is closed, $R_{n+1} \subseteq R_n$, $\Omega|_{R_n}$ is strong, and $T : R_n \rightarrow R_{n+1}$ is a bijective, bounded-both-ways symplectomorphism, with $\|Tx\| \geq c\|x\|$ on R_n using the same constant $c = 1/\|T\|$ as on all of H .*

Proof. By induction on n ; $n = 0$ is Facts 3.2–3.4. Assume the statement at n . Then $R_{n+1} = T(R_n) \subseteq R_n$ (also seen directly, by induction, from $R_1 \subseteq R_0 = H$ and $R_n \subseteq R_{n-1} \Rightarrow T(R_n) \subseteq T(R_{n-1})$). So T restricts to a bounded, $\Omega|_{R_n}$ -preserving self-map of the strong symplectic space $(R_n, \Omega|_{R_n})$. Since $\|T|_{R_n}\| \leq \|T\|$, Fact 3.2’s bound $\|Tx\| \geq \|x\|/\|T|_{R_n}\|$ gives $\|Tx\| \geq c\|x\|$ on R_n ; the closed-range argument gives R_{n+1} closed; the bijectivity argument gives $T : R_n \rightarrow R_{n+1}$ bounded-both-ways; and Fact 3.3 gives $\Omega|_{R_{n+1}}$ strong. $\square$

Define, via Fact 3.4 applied to $(R_n, \Omega|_{R_n})$ and the self-map $T|_{R_n}$,

$$N_n := \{v \in R_n : \Omega(v, R_{n+1}) = 0\}, \quad R_n = R_{n+1} \oplus N_n \quad (\Omega(R_{n+1}, N_n) = 0).$$

Proposition 3.7. (a) $T(N_n) = N_{n+1}$, bijectively and bounded both ways.

(b) $N_n = T^n(N_0)$, and $T^n|_{N_0} : N_0 \rightarrow N_n$ is a bijective, bounded-both-ways, Ω -preserving isomorphism (a symplectomorphism, not a metric isometry in general).

(c) $\Omega(N_j, N_k) = 0$ for all $j \neq k$ (full pairwise Ω -orthogonality).

(d) For every n : $H = R_n \oplus N_{n-1} \oplus \cdots \oplus N_0$ (topological internal direct sum).

Proof. (a) $T(N_n) \subseteq N_{n+1}$: let $v \in N_n$, so $v \in R_n$ and $\Omega(v, R_{n+1}) = 0$. Then $Tv \in T(R_n) = R_{n+1}$. For any $t \in R_{n+2} = T(R_{n+1})$, write $t = Ts'$ with $s' \in R_{n+1}$; then $\Omega(Tv, t) = \Omega(Tv, Ts') = \Omega(v, s') = 0$ (T preserves Ω ; $s' \in R_{n+1}$ and $v \perp_{\Omega} R_{n+1}$). So $\Omega(Tv, R_{n+2}) = 0$, i.e. $Tv \in N_{n+1}$.

$N_{n+1} \subseteq T(N_n)$: let $w \in N_{n+1} \subseteq R_{n+1} = T(R_n)$. By bijectivity of $T : R_n \rightarrow R_{n+1}$ there is a unique $v \in R_n$ with $Tv = w$; write $v = r + \nu$ with $r \in R_{n+1}$, $\nu \in N_n$ (unique decomposition, Fact 3.4 at level n). Then

$$w = Tv = Tr + T\nu, \quad Tr \in T(R_{n+1}) = R_{n+2}, \quad T\nu \in T(N_n) \subseteq N_{n+1}$$

(the last inclusion by the direction just proved). But $R_{n+1} = R_{n+2} \oplus N_{n+1}$ (Fact 3.4 at level $n+1$) is also a unique decomposition, and $w \in N_{n+1}$ means $w = 0 + w$ is *its* decomposition into an R_{n+2} -part (0) and an N_{n+1} -part (w). Comparing forces $Tr = 0$ and $T\nu = w$; since T is injective, $r = 0$, so $v = \nu \in N_n$ and $Tv = w$. Hence $w \in T(N_n)$.

So $T(N_n) = N_{n+1}$; this restriction of the bijection $T : R_n \rightarrow R_{n+1}$ is itself bijective and bounded both ways.

(b) Immediate by induction from (a).

(c) Say $j > k$; then $N_j = T^j(N_0) \subseteq T^j H = R_j \subseteq R_{k+1}$ (decreasing chain, as $j \geq k+1$), and $\Omega(N_k, R_{k+1}) = 0$ by definition of N_k ; hence $\Omega(N_k, N_j) = 0$, and $\Omega(N_j, N_k) = -\Omega(N_k, N_j) = 0$.

(d) Iterate Fact 3.4's decomposition $R_k = R_{k+1} \oplus N_k$ for $k = 0, \dots, n-1$. $\square$

In Example 3.1: $N_n = T^n(\text{first block}) = \{(0, \dots, 0, T_0^n a, 0, \dots) : a \in \mathbb{R}^2\}$ sitting in block $n+1$; part (c) holds trivially since different blocks are already g -orthogonal (hence Ω -orthogonal, as Ω is block-diagonal too).

3.3 The residual space H_∞

$$H_\infty := \bigcap_{n \geq 0} R_n.$$

Theorem 3.8. $T(H_\infty) = H_\infty$, and $T|_{H_\infty}$ is a bounded bijection with bounded inverse — a symplectic automorphism of $(H_\infty, \Omega|_{H_\infty})$, not necessarily metrically unitary.

Proof. $T(H_\infty) = T(\bigcap_n R_n) \subseteq \bigcap_n T(R_n) = \bigcap_n R_{n+1} = H_\infty$. For surjectivity: let $h \in H_\infty \subseteq R_1 = TH$. Because T is globally injective (Fact 3.2), there is a unique $x \in H$ with $Tx = h$. Fix n ; since $h \in R_{n+1} = T(R_n)$, there is $y \in R_n$ with $Ty = h$; by global injectivity $y = x$, so $x \in R_n$. As n was arbitrary, $x \in \bigcap_n R_n = H_\infty$, and $Tx = h$. So $T(H_\infty) = H_\infty$. Boundedness of T^{-1} on H_∞ follows from Fact 3.2's global bound restricted to H_∞ : $\|T^{-1}h\| \leq \|h\|/c$. $\square$

Nothing here used isometry — only injectivity and the bounded-below estimate, both free consequences of Ω -preservation. In Example 3.1, $H_\infty = \bigcap_n R_n = \{0\}$: R_n is exactly “supported in blocks $> n$,” and no nonzero vector is supported past every finite block.

3.4 Where isometry was really needed

Combining §3.2–3.3: for every finite n we have the decomposition (d) of Proposition 3.7, and H_∞ sits inside every R_n . The missing statement, compared to Theorem 2.2, is the infinite one:

$$H \stackrel{?}{=} H_\infty \oplus \left\{ \sum_{n \geq 0} T^n a_n : a_n \in N_0, \sum_n \|a_n\|^2 < \infty \right\}.$$

The isometric proof of this used $\|T^n a\| = \|a\|$ in two places: (i) to get the Pythagorean identity $\|\sum_{n < K} T^n a_n\|^2 = \sum_{n < K} \|a_n\|^2$, which needs the pieces to be g -orthogonal, not just Ω -orthogonal; and (ii) to make the resulting map $\Psi : \ell^2(N_0) \rightarrow H$ bounded. Without isometry we only have

$$c^n \|a\| \leq \|T^n a\| \leq \|T\|^n \|a\| \quad (a \in N_0),$$

and no control on $g(T^n a, T^m b)$ for $a, b \in N_0$, $n \neq m$ — only on $\Omega(T^n a, T^m b)$ (Prop. 3.7(c)). So $\sum_n T^n a_n$ need not converge for a general square-summable (a_n) .

Example 3.9 (Concrete failure). Take $a_n := (1/n)e_1 \in N_0 \cong \mathbb{R}^2$ ($n \geq 1$; a genuinely ℓ^2 sequence). Then $T^n a_n$ lives in block $n+1$ with value $T_0^n a_n = (\lambda^n/n)e_1$, so $\|T^n a_n\| = \lambda^n/n \rightarrow \infty$: the partial sums $\sum_{n \leq K} T^n a_n$ are not even bounded, let alone convergent.

3.5 Recovering an honest infinite decomposition: go metric, not symplectic

The fix (Richter, Shimorin, and — closest to what we need — Manolescu [2]) is to build the wandering space out of the *metric* (g -)orthogonal complement instead of the symplectic one, because g -orthogonal projections are automatically bounded by 1 regardless of how much T distorts norms.

Left invertibility. T bounded below (Fact 3.2) $\iff T^*T$ invertible $\iff T$ has a bounded left inverse $T^- := (T^*T)^{-1}T^*$ (so $T^-T = I$); a general Hilbert-space fact, here a consequence of Ω -preservation.

Metric wandering space. $W := H \ominus TH = \ker T^*$. By Lemma 1.2 (no isometry needed),

$$N_0 = R_1^{\perp \Omega} = J(R_1^{\perp g}) = J(W).$$

So the base-level relation between the metric and symplectic wandering spaces is unconditional: $N_0 = J(W)$, always. (At higher levels $N_n = T^n(N_0)$ and $T^n W$ need not correspond simply through J unless $TJ = JT$ on the relevant subspaces.)

The class $\mathcal{D}$ condition. Following Manolescu (adapting Shimorin [1]), say $T \in \mathcal{D}$ if

$$(T^n)^- = (T^-)^n \quad \text{for all } n \geq 2.$$

This holds automatically if T is an isometry ($T^- = T^*$, trivially); it also holds for every bounded-below weighted shift with weights bounded away from 0 and ∞ , and more generally for direct sums of operators in $\mathcal{D}$ acting on g -orthogonal summands.

Theorem 3.10 (Manolescu 2022, Thm. 2.9–2.10, restated). *If $T \in \mathcal{D}$, then $P_n := T^n(T^-)^n$ is the g -orthogonal projection onto $R_n = T^n H$; P_n converges strongly to the orthogonal projection P_∞ onto $H_\infty = \bigcap_n R_n$; and*

$$H = H_\infty \oplus_g H_s, \quad H_s := \overline{\bigoplus_{n \geq 0} T^n W},$$

a genuine (g -)orthogonal Hilbert-space direct sum, with H_∞, H_s reducing T and $T|_{H_\infty}$ bijective. In general $T|_{H_s}$ is only a bounded-below, bounded-above per-summand map $T^n W \rightarrow T^{n+1} W$ — a weighted shift, not literally conjugate to $\Sigma \otimes I_W$ via an isometry.

Example 3.11 (Example 3.1 is in $\mathcal{D}$). In standard per-block coordinates $x = (x^{(1)}, x^{(2)})$, the diagonal $T_0 = \text{diag}(\lambda, \lambda^{-1})$ makes $H = \bigoplus_n \mathbb{R}^2$ split g -orthogonally as $H = \ell_{(1)}^2 \oplus_g \ell_{(2)}^2$, and $T = T^{(1)} \oplus T^{(2)}$ where $T^{(i)}$ is the scalar weighted shift with constant weight λ (resp. λ^{-1}). Each constant-weight scalar shift is in $\mathcal{D}$ (direct computation), hence so is their g -orthogonal direct sum T . Theorem 3.10 then gives $H = H_\infty \oplus_g H_s$ with $H_\infty = \{0\}$, so $H_s = H$: the exhaustion by full blocks $T^n W$ is genuinely orthogonal, even though the per-summand map $T^n W \rightarrow T^{n+1} W$ scales the two eigen-components by wildly different factors ($\lambda^{\pm 1}$ each step).

3.6 The Toeplitz / J -shift symbol still makes sense

Nothing in Lemma 2.3 used isometry — only Ω -preservation, iterated. So, verbatim: for $a, b \in N_0$ and $k \geq 0$, set $\varphi_k(a, b) := \Omega(T^k a, b) = g(JT^k a, b)$, $\Phi_k := P_{N_0} J T^k|_{N_0}$; then for $n \geq m$: $\Omega(T^n a, T^m b) = \varphi_{n-m}(a, b)$, and for finite combinations this reproduces the same block-Toeplitz expression as Theorem 2.4, term by term, with the weaker bound $\|\Phi_k\| \leq \|J\| \|T^k|_{N_0}\| \leq \|T\|^k$ (instead of ≤ 1). In Example 3.1, the identical computation gives $\Phi_0 = J_0$, $\Phi_k = 0$ ($k \geq 1$), regardless of λ : the symbol does not “see” the non-isometry here, because the distortion acts entirely within each Ω -orthogonal block.

3.7 Summary

	T isometric	T only Ω -preserving
Bounded below, closed range, $H = R \oplus N_0$	unconditional	unconditional
Ω -filtration $N_n = T^n N_0$, pairwise $\Omega(N_j, N_k) = 0$	yes	unconditional (§3.2)
H_∞ : T bijective, bdd. inverse there	yes	unconditional (§3.3)
Reduction formula $\Omega(T^n a, T^m b) = \varphi_{n-m}(a, b)$	yes	unconditional (§3.6)
$H = H_\infty \oplus_g H_s$, convergent ℓ^2 -sum, $T _{H_s} \cong \Sigma \otimes I$	automatic	only if $T \in \mathcal{D}$; then g - (not Ω -)orthogonal pieces, $T _{H_s}$ a weighted shift
Symbol bound $\ \Phi_k\ \leq 1$	yes	no ($\leq \ T\ ^k$ only)

References

- [1] S. Shimorin, *Wold-type decompositions and wandering subspaces for operators close to isometries*, J. Reine Angew. Math. **531** (2001), 147–189.
- [2] L. Manolescu, *Wold decomposition for operators close [to] isometries*, Sci. Bull. Politeh. Univ. Timișoara, Trans. Math. Phys. (2022).